

Execution-Aware Portfolio Reinforcement Learning via Proximal Execution Control

Separating Target and Executed Portfolios for Cost-Aligned Evaluation and Execution Stabilization

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Abstract

Most portfolio reinforcement learning (RL) formulations treat the portfolio weights output by the policy as the portfolio that is immediately executed. Under transaction costs, this interface conflates target decisions with realized holdings and obscures the object on which trading costs are actually incurred. We separate these objects through a two-stage decomposition: the policy outputs a target portfolio w_t^{tgt} , an execution controller maps it to an executed portfolio w_t^{exec} , and both transaction costs and primary evaluation are attached to executed turnover on the realized path. We study this interface in a frozen-policy, validation-first protocol that reuses one learned policy across execution arms and varies only the execution rate η . On a fixed 27-name U.S. large-cap universe, the validation-selected interior point $\eta = 0.5$ roughly halves executed turnover and improves net Sharpe in positive-cost regimes, while the $\kappa = 0$ effect is negligible. The gain is explained by lower realized turnover and lower realized cost rather than by higher gross returns, and auxiliary retraining, second-universe, and cost-calibrated linear-convex comparator checks all point in the same friction-sensitive direction.

Keywords: portfolio RL, transaction costs, execution, turnover, proximal control, reproducible backtesting

1 Introduction

Portfolio RL is often formulated as if the portfolio weights output by the policy are the portfolio that is immediately held. This direct-action interface is convenient, but under transaction costs it collapses two different objects into one: the target portfolio proposed at decision time and the executed portfolio that is actually carried into the next return interval.

When execution is gradual, realized turnover, realized cost, and realized net return are determined by the executed path rather than by the raw target action alone. The distinction therefore matters even before one asks whether a different policy should be learned. It changes which portfolio is charged for trading costs, which return path is evaluated, and how one should interpret the relation between portfolio choice and trade implementation.

This paper studies a focused identification question: under a fixed learned policy, does separating target and executed portfolios create an interior execution frontier in positive-cost regimes, and are the resulting net-performance changes explained by lower realized turnover and lower realized cost rather than by improved gross returns?

We adopt a frozen-policy protocol because retraining would change both the target generator and the execution interface at the same time. That joint comparison is important, but it answers a different question. Holding the learned policy fixed and varying only the execution mapping isolates how execution-aware accounting and partial execution affect realized net performance for the same learned target path.

Contributions.

1. **Execution-aware portfolio interface and accounting.** We separate target and executed portfolios and attach transaction costs and primary evaluation to the realized executed path.

2. **Frozen-policy, validation-first identification protocol.**

We evaluate one learned policy across fixed execution arms, select the operating point on validation only, and reserve the test window for held-out evaluation so that execution effects are separated from retraining effects.

3. **Friction-sensitive empirical evidence and scope checks.**

On the fixed 27-name U.S. large-cap benchmark, interior execution rates reduce turnover and improve net Sharpe in positive-cost regimes; η -aligned retraining, a second 36-name benchmark, and a cost-calibrated linear-convex information-parity comparator all point in the same implementation-driven direction.

The contribution is therefore not a stronger target generator, but an execution-aware accounting study of how partial execution changes realized net performance under frictions for a fixed learned policy.

2 Related Work

Classical friction-aware portfolio choice. Classical portfolio theory studies dynamic allocation under risk/return trade-offs [1, 2]. Once trading frictions are introduced, however, the problem is no longer only which portfolio is desirable in isolation, but how current holdings should be adjusted toward that portfolio over time. Friction-aware portfolio optimization therefore shifts attention from static allocation to dynamic adjustment under transaction costs [9, 11]. In practice, that literature also motivates turnover-aware convex updates and linear prediction-to-portfolio mappings, which provide a natural non-RL comparison class for the execution-aware accounting layer studied here.

Dynamic trading, aim portfolios, and partial adjustment.

A closer conceptual predecessor is the dynamic-trading literature that separates current holdings from desired holdings and studies how portfolios move toward an aim portfolio under costs. Gârleanu and Pedersen make that distinction explicit in a dynamic setting with predictable returns and transaction costs [11]. This literature also motivates partial adjustment and inaction-region reasoning once trading is costly. Our paper adopts the same basic separation, but relocates it into the portfolio-RL interface: the contribution is not a new frictional trading concept, but a re-specification of the accounting and evaluation object inside RL backtesting, where a learned target portfolio and the realized executed portfolio should not be conflated.

Optimal execution. Optimal-execution models also separate a desired position from the path used to reach it. Bertsimas and Lo and Almgren and Chriss study gradual trading that balances execution cost and risk, typically given an inventory objective or liquidation target [3, 12]. Our η -update is similar in implementation spirit to partial-adjustment rules from that literature, but the setting is different. We do not study an intraday execution optimizer. Instead, we study a daily execution layer that maps portfolio-policy outputs to executed holdings and changes how turnover, cost, and net performance are recorded for a fixed target path.

Portfolio RL, smoothing, and regularization. Most portfolio RL and deep portfolio management papers use a direct-action interface in which the policy output is treated as the portfolio that is immediately held or rebalanced to [4, 5, 7]. Even when transaction costs are included, turnover is often computed directly from consecutive actions, so action turnover becomes a proxy for executed turnover. Related turnover-aware sequential-allocation work outside RL likewise studies frictions at the level of portfolio updates [10]. Recent portfolio-RL variants improve robustness through richer state construction, auxiliary prediction, or reward shaping [7, 8], and nearby RL work studies execution agents, hierarchical decomposition, or smooth-action regularization [6, 13, 14, 15]. The distinction in the present paper is therefore not simply “smoother actions.” It is that executed holdings are treated as the first-class accounting and evaluation object.

Taken together, the prior literature already contains gradual adjustment, aim portfolios, execution scheduling, and smoothing ideas. What is less explicitly isolated is a portfolio-RL experimental protocol in which target and executed portfolios are separated, primary evaluation is attached to the realized executed path, and the empirical effect of that separation is studied under a frozen learned policy. That focused identification question is the one addressed here.

3 Problem Formulation

We study an episodic daily portfolio-control problem over N risky assets. The action space is the long-only fully invested simplex, there is no explicit cash asset, decisions are made once per trading day, and timing follows a close-to-close convention. At decision time t , the agent may use only information available up to close t ; the portfolio held over $(t, t+1]$ is the executed portfolio formed after the decision at t .

We denote adjusted close prices by $p_t \in \mathbb{R}^N$. Single-period arithmetic returns over interval $(t, t+1]$ are

$$r_{t+1} \triangleq \frac{p_{t+1} - p_t}{p_t} \in \mathbb{R}^N.$$

3.1 State

At decision time t , the agent observes state s_t constructed only from information available up to time t . In the empirical instantiation, s_t contains a rolling return window, per-asset volatility estimates, previous executed weights, and optional frozen signal channels; Section 6 gives the exact window lengths and feature counts.

3.2 Action

The policy outputs a target portfolio $w_t^{\text{tgt}} = \pi_\theta(s_t)$ in the long-only fully invested simplex

$$\Delta^N \triangleq \{w \in \mathbb{R}_+^N : \mathbf{1}^\top w = 1\}.$$

Thus the MDP action is a *target* portfolio rather than the realized holding vector. In the empirical implementation, the actor emits unconstrained real-valued logits that are mapped into Δ^N by a stable softmax; Section 6 records that concrete parameterization.

3.3 Execution Mapping

Let the previous executed portfolio be w_{t-1}^{exec} . The execution layer maps the target portfolio to an executed portfolio according to

$$w_t^{\text{tgt}} = \pi_\theta(s_t) \in \Delta^N, \quad (1)$$

$$w_t^{\text{exec}} = (1 - \eta_t)w_{t-1}^{\text{exec}} + \eta_t w_t^{\text{tgt}}, \quad \eta_t \in (0, 1], \quad (2)$$

$$\text{TO}_t^{\text{tgt}} = \|w_t^{\text{tgt}} - w_{t-1}^{\text{exec}}\|_1, \quad (3)$$

$$\text{TO}_t^{\text{exec}} = \|w_t^{\text{exec}} - w_{t-1}^{\text{exec}}\|_1. \quad (4)$$

Here TO_t^{tgt} is the turnover implied by an immediate move to the target portfolio, while $\text{TO}_t^{\text{exec}}$ is the turnover actually realized on the executed path.

3.4 Reward and Realized Return

Transaction costs are charged on executed turnover, and realized reward is attached to the executed portfolio:

$$\text{cost}_t = \kappa \text{TO}_t^{\text{exec}}, \quad (5)$$

$$R_{t+1} = (w_t^{\text{exec}})^\top r_{t+1} - \text{cost}_t, \quad (6)$$

$$R_{t+1}^{\text{tgt}} = (w_t^{\text{tgt}})^\top r_{t+1} - \kappa \text{TO}_t^{\text{tgt}}. \quad (7)$$

The quantity R_{t+1} is the realized executed-path return used for primary evaluation. The quantity R_{t+1}^{tgt} is a hypothetical target-path return used only for diagnostics. The MDP action is therefore w_t^{tgt} , while realized portfolio dynamics and primary reward are governed by the executed path w_t^{exec} and (6).

3.5 Terminology and Evaluation Objects

Throughout the paper, *target portfolio* denotes the policy output w_t^{tgt} , *executed portfolio* denotes the realized holdings w_t^{exec} , *immediate-execution baseline* denotes the comparison arm $\eta = 1.0$, *validation-selected operating point* denotes the execution rate chosen on the validation window, and *net Sharpe* denotes the annualized Sharpe ratio computed from executed net linear returns. The code label `sharpe_net_lin` is the implementation name for this same net-Sharpe quantity.

Table 1: Core versus diagnostic quantities.

Quantity	Role	Primary evaluation
TO_t^{tgt}	target-turnover diagnostic	No
$\text{TO}_t^{\text{exec}}$	executed turnover	Yes
R_{t+1}^{tgt}	hypothetical target-path return	No
R_{t+1}	realized executed-path return	Yes
$\ w_t^{\text{exec}} - w_t^{\text{tgt}}\ _2$	tracking diagnostic	No

Primary portfolio metrics such as Sharpe, CAGR, and max drawdown are computed from the realized executed-path returns $\{R_{t+1}\}$. Diagnostics may additionally use w_t^{tgt} , TO_t^{tgt} , R_{t+1}^{tgt} , or target-executed tracking discrepancies, but these objects are explicitly labeled and never serve as the primary evaluation metric.

4 Execution-Aware Control and Accounting

This section defines the execution-aware interface and the frozen-policy accounting protocol used throughout the experiments. The theoretical section that follows is narrower: it records simple analytical properties of the execution map that help interpret the empirical results without overstating the depth of the theory contribution.

4.1 Two-stage control

We interpret portfolio RL under frictions as a two-stage control problem:

1. **Policy layer:** the agent chooses a desired allocation $w_t^{\text{tgt}} = \pi_\theta(s_t)$.
2. **Execution layer:** the controller maps that desired allocation into realized holdings w_t^{exec} via (2).

The formulation naturally supports learning under the executed dynamics (6). In the present empirical study, however, the policy layer is trained exactly once and then held fixed; every empirical arm reuses that same trained policy and varies only the execution layer. The resulting comparison isolates how execution-aware accounting changes realized performance for the same learned target path.

4.2 Fixed execution timescale

The execution update (2) is fixed, and the present paper studies only constant timescales $\eta_t \equiv \eta$. This is an identification choice rather than only a simplification. Once η_t becomes rule-based or learned, the object of study shifts from execution accounting to adaptive controller design. Fixed η removes

schedule-design confounds, preserves a one-parameter execution frontier, and keeps the empirical question at the level of how partial execution alone changes realized turnover, realized cost, and net performance.

4.3 Cost alignment (core principle)

Cost should be attached to what is actually traded, not to what was merely proposed. The primary accounting object is therefore executed turnover $\text{TO}_t^{\text{exec}}$ and the realized executed-path return (6). Target-based cost and target-path return remain useful as counterfactual diagnostics, but they are not the primary accounting object because they describe a hypothetical immediate move that was not actually executed. Charging costs on target turnover TO_t^{tgt} induces systematic friction mis-scaling when $\eta_t < 1$ (Proposition 2); the present paper studies that accounting implication under a frozen-policy protocol rather than running alternative retraining experiments.

5 Theoretical Analysis

This section collects simple analytical properties needed for interpretation: the proximal form of the update, realized-turnover scaling, tracking and low-pass behavior, and the accounting distortion induced by target-based cost charging.

Proposition 1 (Proximal interpretation). *For constant $\eta \in (0, 1)$, the execution update (2) is the unique minimizer of:*

$$w_t^{\text{exec}} = \arg \min_{w \in \Delta^N} \frac{1}{2} \|w - w_{t-1}^{\text{exec}}\|_2^2 + \frac{\eta}{2(1-\eta)} \|w - w_t^{\text{tgt}}\|_2^2.$$

Proof sketch. The objective is strongly convex in w . Setting the gradient to zero yields a convex combination of w_{t-1}^{exec} and w_t^{tgt} ; the simplex constraint is preserved by convexity of Δ^N . Since both w_t^{tgt} and w_{t-1}^{exec} lie in the simplex, their convex combination is naturally feasible without further projection. $\square$

Remark 1. *Execution is a proximal step that penalizes changes from the previous executed portfolio, regularizing the trading path while still tracking the target. That same convex-combination form also makes the rule computationally light and feasibility-preserving under the simplex constraint.*

Lemma 1 (Turnover scaling and cumulative bound). *Under (2)–(4), for any $\eta_t \in (0, 1]$:*

$$\text{TO}_t^{\text{exec}} = \eta_t \text{TO}_t^{\text{tgt}}.$$

If additionally $\eta_t \leq \eta_{\max}$ for all t , then

$$\sum_{t=1}^T \text{TO}_t^{\text{exec}} \leq \eta_{\max} \sum_{t=1}^T \text{TO}_t^{\text{tgt}}.$$

Proof sketch. From (2), $w_t^{\text{exec}} - w_{t-1}^{\text{exec}} = \eta_t(w_t^{\text{tgt}} - w_{t-1}^{\text{exec}})$. Taking ℓ_1 norms gives the one-step identity. Summing over t and using $\eta_t \leq \eta_{\max}$ yields the cumulative bound. $\square$

Lemma 2 (Tracking and low-pass behavior). *Define instantaneous tracking discrepancy by $\|w_t^{\text{exec}} - w_t^{\text{tgt}}\|_2$. Then*

$$w_t^{\text{exec}} - w_t^{\text{tgt}} = (1 - \eta_t)(w_{t-1}^{\text{exec}} - w_t^{\text{tgt}}),$$

so for fixed $(w_{t-1}^{\text{exec}}, w_t^{\text{tgt}})$, smaller η_t increases discrepancy whenever $w_{t-1}^{\text{exec}} \neq w_t^{\text{tgt}}$. For constant $\eta \in (0, 1]$, the executed sequence has the closed form:

$$w_t^{\text{exec}} = (1 - \eta)^t w_0^{\text{exec}} + \eta \sum_{i=1}^t (1 - \eta)^{t-i} w_i^{\text{tgt}}.$$

For fixed horizon t , $\eta \rightarrow 0$ implies $w_t^{\text{exec}} \rightarrow w_0^{\text{exec}}$ (anchor convergence).

Proof sketch. Rearranging (2) yields the tracking identity. Unrolling the constant- η recursion yields the closed form, which is an exponential smoothing of the target path. $\square$

Proposition 2 (Cost misalignment rescaling). *If one incorrectly charges costs using target turnover TO_t^{tgt} while execution obeys (2), then the charged cost overestimates realized cost by factor $1/\eta_t$:*

$$\kappa \text{TO}_t^{\text{tgt}} = \frac{\kappa}{\eta_t} \text{TO}_t^{\text{exec}}.$$

Equivalently, the target-based cost distortion is

$$\begin{aligned} \kappa \text{TO}_t^{\text{tgt}} - \kappa \text{TO}_t^{\text{exec}} &= \kappa (\text{TO}_t^{\text{tgt}} - \text{TO}_t^{\text{exec}}) \\ &= \kappa \left(\frac{1}{\eta_t} - 1 \right) \text{TO}_t^{\text{exec}}. \end{aligned}$$

Proof sketch. Substitute Lemma 1 into $\kappa \text{TO}_t^{\text{tgt}}$ and subtract the executed cost term. $\square$

Interpretation. Target-based charging inflates the effective friction scale whenever $\eta_t < 1$. If that target-based cost is inserted into training rewards, the friction coefficient appearing in critic targets and policy updates is larger than the one actually realized on the executed path. The present paper does not quantify full policy-gradient bias or retrain under alternative rewards; it records the accounting distortion and its direction.

These properties are simple but useful: they clarify what quantity is actually being regularized, what turnover is actually realized, and how target-based accounting distorts the friction scale when $\eta < 1$.

6 Experimental Setup

6.1 Data, universe, and splits

We use a fixed 27-name large-cap U.S. equity snapshot with daily adjusted close prices retrieved via `yfinance`. The frozen-control study uses:

- Train window: 2010-01-01 to 2021-12-31.
- Validation window: 2022-01-01 to 2023-12-31.
- Test window: 2024-01-01 to 2025-12-31.
- Decision frequency: daily rebalancing.

Because the state uses 30-day rolling features, the realized validation trace spans 2022-02-15 to 2023-12-29 and the realized held-out test trace spans 2024-02-14 to 2025-12-31. Matched external baselines are evaluated on these same effective held-out dates. Timing follows a close-to-close convention: features at time t use information available by close t , the execution update forms w_t^{exec} after the decision at t , and

realized portfolio return uses r_{t+1} over $(t, t + 1]$. This fixed 27-name large-cap U.S. equity snapshot is reproducible but not a point-in-time constituent reconstruction, so the empirical claims should be read as controlled benchmark evidence rather than as a survivorship-bias-free deployment backtest. Survivorship-related bias may affect the absolute performance level. Because all η -arms share the same universe, dates, and frozen target policy, the within-policy execution comparison is less sensitive to this bias than absolute cross-strategy comparisons, though it is not immune. Internal 2026 forward checks were used for engineering sanity only and are not part of the main paper tables. Appendix B collects the data and protocol details in one place for auditability.

6.2 Observations and environment

The empirical instantiation uses a 30-day return window ($L = 30$), a 30-day volatility window ($L_v = 30$), previous executed weights w_{t-1}^{exec} , and two frozen signal channels: `reversal_5d` and `short_term_reversal`. The return block stores flattened per-asset log returns computed from adjusted-close ratios. The volatility block stores the 30-day rolling standard deviation (`ddof = 0`) of those same log returns. The previous-weight block stores the simplex-valued executed holdings from time $t - 1$, and the signal channels are appended in the same asset order. For the fixed 27-asset universe, this yields $30 \times 27 + 27 + 27 + 2 \times 27 = 918$ observation dimensions, matching the stored run metadata.

No additional online whitening or environment-level normalization is applied beyond the upstream feature construction. Realized portfolio return is evaluated on arithmetic returns obtained from the stored log-return frame, and anti-lookahead is enforced by forming s_t only from information available up to time t and applying w_t^{exec} to interval return r_{t+1} .

The actor outputs a 27-dimensional real-valued action vector. In the environment, that vector is clipped elementwise to $[-1, 1]$ and mapped to long-only fully invested target weights by a stable softmax with logit scale 10.0. This implementation detail matters because it makes the simplex constraint operational: the learned action parameterization always produces a valid target portfolio before the execution layer is applied.

6.3 Frozen controller and execution arms

- **PRL-SAC control model:** a Stable-Baselines3 SAC controller with an `MlpPolicy` actor-critic backbone and volatility-conditioned PRL plasticity schedule. For each seed, the controller is trained once on the training window, producing one frozen checkpoint. Evaluation then reuses that same checkpoint across all η -arms for that seed. The policy outputs target-allocation logits; the environment maps them to simplex weights before the execution layer applies partial adjustment. The exact training hyperparameters and frozen scheduler settings are summarized in Appendix C.
- **Execution-control arms:** the same trained policy is evaluated under the *a priori* fixed grid $\eta \in \{1.0, 0.5, 0.2, 0.1, 0.082, 0.05, 0.02\}$, with $\eta = 1.0$ as the immediate-execution baseline.

6.4 Internal comparison arms

The main comparison in this paper is internal and isolates execution while holding the learned controller fixed:

- **Immediate-execution baseline:** fixed $\eta = 1.0$.
- **Execution-aware variants:** the interior η values from that same fixed grid.

This design keeps the predictive model constant and attributes performance differences to execution alone. Matched external heuristics are handled separately so that the main causal comparison remains “same frozen policy, different execution rule,” while non-RL reference portfolios are added only under the same accounting conventions.

6.5 Transaction cost regimes

Sweep $\kappa \in \{0, 5 \times 10^{-4}, 10^{-3}\}$ and evaluate the fixed η grid with paired seeds.

6.6 A priori η grid and validation-only selection

The η grid is fixed before evaluation and is not adapted after looking at test outcomes. It combines the immediate-execution boundary $\eta = 1.0$, coarse interior values $\{0.5, 0.2, 0.1\}$, and finer low- η probes $\{0.082, 0.05, 0.02\}$ to inspect frontier curvature in the low-adjustment region. The value $\eta = 0.082$ is included because it is the incumbent `rebalance_eta` of the frozen control snapshot used to train the reference policy; it is therefore a pre-existing control operating point rather than a post-hoc addition after inspecting validation or test performance. Let the positive-cost validation set be $\mathcal{K}^+ = \{5 \times 10^{-4}, 10^{-3}\}$. For each η in the fixed grid, we define the validation score

$$\text{Score}(\eta) \triangleq \frac{1}{|\mathcal{K}^+|} \sum_{\kappa \in \mathcal{K}^+} \text{median}_{s \in \mathcal{S}} \text{Sharpe}_{\kappa, \eta, s}^{\text{net}, \text{lin}}.$$

We then select the *largest* η satisfying

$$\text{Score}(\eta) \geq \tau(b), \quad b \triangleq \max_{\eta'} \text{Score}(\eta'),$$

with the signed relative threshold

$$\tau(b) \triangleq \begin{cases} 0.95b, & b > 0, \\ b/0.95, & b < 0, \\ 0, & b = 0. \end{cases}$$

Thus, η selection is based on the validation window only and favors the least aggressive execution smoothing among near-best validation operating points. The 2024–2025 test window is used only for final held-out evaluation of the selected operating point and is never used for η selection.

6.7 Evaluation protocol (paired seeds)

Let $\mathcal{S} = \{0, 1, \dots, 9\}$. For each seed $s \in \mathcal{S}$, evaluate all arms (different η values) with identical data splits and the same frozen trained policy per seed. Report:

- Median across seeds for each reported metric in the main tables.

- Paired differences (e.g., ΔSharpe) and win-rate (% seeds improved) versus the immediate-execution baseline $\eta = 1.0$.
- For the selected- η vs. immediate-execution comparison, also report paired dispersion (IQR), exact one-sided sign-test p -values, two-sided Wilcoxon signed-rank p -values, and percentile-bootstrap 95% confidence intervals for the paired median ΔSharpe and ΔCAGR .

These seed-level tests quantify optimization-level variability conditional on a fixed historical window, not independent market-sample uncertainty.

6.8 Matched heuristic context

To provide limited non-RL context without changing the main question, the rebuilt protocol also includes five heuristic baselines on the same effective held-out dates within the test split: buy-and-hold equal weight, daily-rebalanced equal weight, inverse-volatility risk parity, minimum-variance, and a simple long-only mean-variance heuristic. The two optimization-style heuristics use a trailing 252-day window with a minimum 30-day warm-up, and the mean-variance heuristic fixes risk aversion at 10.0. These baselines are not allowed to use a different accounting convention. They must use the same transaction-cost regime κ , the same realized-path net-return definition, the same Sharpe annualization $\sqrt{252}$, the same risk-free assumption $r_f = 0$, the same executed-path primary metric definitions as the RL arms, and the same effective evaluation dates induced by the frozen split and rolling-feature availability. They are included only to calibrate external performance scale under aligned accounting. They are not part of the identification strategy, and they are not used for selection, validation, or the paper’s core causal interpretation.

6.9 CC-TA-LBIP baseline

As a stronger non-RL comparator, we also include a *Cost-Calibrated Turnover-Aware Linear Information-Parity Baseline* (CC-TA-LBIP). CC-TA-LBIP uses the same 918-dimensional state as the RL controller—the 30-day return window, the current volatility vector, the previous executed weights, and the same two frozen signal channels—but replaces the nonlinear policy map with per-asset ridge regressions that produce a linear expected-return estimate $\hat{\mu}_t$. Those linear predictions are then passed to a long-only convex optimizer with an explicit turnover anchor,

$$\max_{w \in \Delta} \hat{\mu}_t^\top w - \frac{\gamma}{2} w^\top \hat{\Sigma}_t w - \frac{c\kappa}{2} \|w - w_{t-1}^{\text{exec}}\|_2^2.$$

Unlike the main RL experiment, this auxiliary comparator does not add a second PEC layer on top of the target weights. The turnover penalty is internalized directly inside the classical optimizer, and the resulting portfolio is executed immediately.

The accounting rules are still matched to the RL experiments. CC-TA-LBIP is evaluated under the same proportional cost grid, the same executed-path net-return definition, and the same annualization convention as the RL arms. To keep this comparator narrow and interpretable, we fix the linear forecast map at `ridge_alpha = 30` and tune only the cost-scaling constant on validation. A small validation-only development

grid over $c \in \{0, 2000, 3000, 4000, 6000, 8000, 12000\}$ selects $c = 3000$ under the same conservative “near-best then least aggressive” logic used elsewhere in the paper: we take the smallest positive c within 95% of the best positive-cost validation score. Because the integrated turnover anchor is proportional to κ , the comparator collapses exactly to the unregularized $c = 0$ baseline when $\kappa = 0$, which avoids any free zero-cost smoothing effect. This comparator remains auxiliary. It broadens the external baseline set under information parity, but it is not part of the frozen-policy identification strategy.

6.10 Metrics

Core metrics (exec only). Sharpe, CAGR, max drawdown, average executed turnover $\overline{\text{TO}}^{\text{exec}}$, and realized cost are computed from the executed-path quantities in Table 1. Specify annualization and risk-free assumption:

- Sharpe annualization: $\sqrt{252}$.
- Risk-free rate: 0.

These same definitions are used for both RL arms and external heuristic baselines.

Diagnostics (target used for analysis only).

- Target turnover $\overline{\text{TO}}^{\text{tgt}}$.
- Tracking error / misalignment measures (defined below).

Diagnostic path comparisons. Using the diagnostic target-path return R_{t+1}^{tgt} from Section 3, we compute path-space comparisons between the realized executed path and the hypothetical immediate target path. A signed mean-gap diagnostic can then be written as

$$\Delta_{\text{mis}} \triangleq \frac{1}{T} \sum_{t=0}^{T-1} (R_{t+1} - R_{t+1}^{\text{tgt}}).$$

In our runs, this signed average is numerically small and less informative than path-space quantities. The main paper therefore emphasizes trace-based absolute diagnostics:

- mean absolute return gap $\frac{1}{T} \sum_t |R_{t+1} - R_{t+1}^{\text{tgt}}|$,
- final equity gap $|E_T^{\text{exec}} - E_T^{\text{tgt}}|$,
- maximum daily absolute gap $\max_t |R_{t+1} - R_{t+1}^{\text{tgt}}|$,
- average weight-path tracking discrepancy.

7 Results

7.1 Protocol framing: validation selection, then held-out evaluation

The empirical protocol is two-stage. First, the fixed η grid is scored on the validation window and one operating point is selected using the rule from Section 6.6. Second, the held-out test window is used only for final evaluation of that validation-selected operating point and the immediate-execution baseline. No test statistic is used to choose η . Every result in this section

reuses the same trained policy and changes only the execution mapping applied to that frozen target path. The main paper evidence is therefore organized around: (i) a validation frontier plus selection report, (ii) a held-out selected- η vs. $\eta = 1.0$ table, (iii) trace-based mechanism diagnostics, (iv) a rolling-window frontier robustness diagnostic, and (v) a denser canonical friction-grid diagnostic. We then add two scope checks: an η -aligned retraining comparison on the canonical U27 benchmark and a second-universe replication on a nearby 36-name large-cap benchmark. The full held-out η frontier is still useful, but only as a post-selection diagnostic rather than as model-selection evidence. Matched heuristics are reported in the appendix for scale only. In the rolling-window extension reported later in this section, the distinction between selected-point transfer and frontier robustness becomes especially important: the selected operating point is window-sensitive, whereas the positive-cost held-out frontier itself is more stable.

7.2 Validation selection locks the operating point at $\eta = 0.5$

Table 2 reports the validation-only selection object. All positive-cost validation scores are positive in this rebuilt run, so the selection problem is about taking the largest η that remains close to the best validation frontier point. The raw best validation score is attained at $\eta = 0.02$ with 0.5213, but the locked 95% rule qualifies six interior grid points and then selects the *largest* qualifying value. The resulting validation-selected operating point is therefore $\eta = 0.5$, while the immediate-execution baseline $\eta = 1.0$ narrowly fails the qualification threshold.

This matters for interpretation. The main empirical claim is no longer that one hand-picked small η looks attractive on test, but that a validation-selected interior operating point transfers to the held-out window without using test outcomes for model selection. Because the trained policy is held fixed throughout, this is an execution-choice result rather than a retraining result. A small threshold-sensitivity check also supports the rule itself: tightening the relative threshold from 95% to 97.5% still selects $\eta = 0.5$, whereas only the looser 90% rule reverts to $\eta = 1.0$. Relative to the raw best point $\eta = 0.02$, the locked choice $\eta = 0.5$ retains 97.66% of the best validation score while keeping average positive-cost tracking error much smaller (0.00257 versus 0.00692). Validation also already shows the turnover mechanism clearly: seed-median executed turnover falls from 0.02160 at $\eta = 1.0$ to 0.01087 at $\eta = 0.5$, a reduction of about 49.7%. Figure 1 visualizes this validation result.

7.3 Held-out test: selected $\eta = 0.5$ vs. immediate execution

Table 3 gives the held-out paired comparison between the validation-selected operating point and the immediate-execution baseline. The main evidence is again concentrated in the positive-cost regimes. At $\kappa = 5 \times 10^{-4}$ and $\kappa = 10^{-3}$, the selected operating point improves paired median net Sharpe by +0.0105 and +0.0213, respectively, and wins on all 10/10 paired seeds in both cases (exact one-sided sign test $p = 9.77 \times 10^{-4}$ for each regime). The correspond-

Table 2: Validation-only η selection on the 2022–2023 split. Let $\kappa^+ = \{5 \times 10^{-4}, 10^{-3}\}$. The score is the mean, over $\kappa \in \kappa^+$, of the seed-median net Sharpe. The selected operating point is the largest η within the locked relative threshold of the best validation score.

η	Score	Avg. paired-median ΔSharpe ($\kappa > 0$)	Avg. seed-median executed turnover ($\kappa > 0$)	Qualifies	Selected
0.500	0.50907	0.01409	0.01087	Yes	Yes
0.200	0.51359	0.02290	0.00501	Yes	No
0.100	0.51704	0.02678	0.00271	Yes	No
0.082	0.51791	0.02768	0.00226	Yes	No
0.050	0.51958	0.02919	0.00143	Yes	No
0.020	0.52126	0.03093	0.00060	Yes	No
1.000	0.49327	0.00000	0.02160	No	No

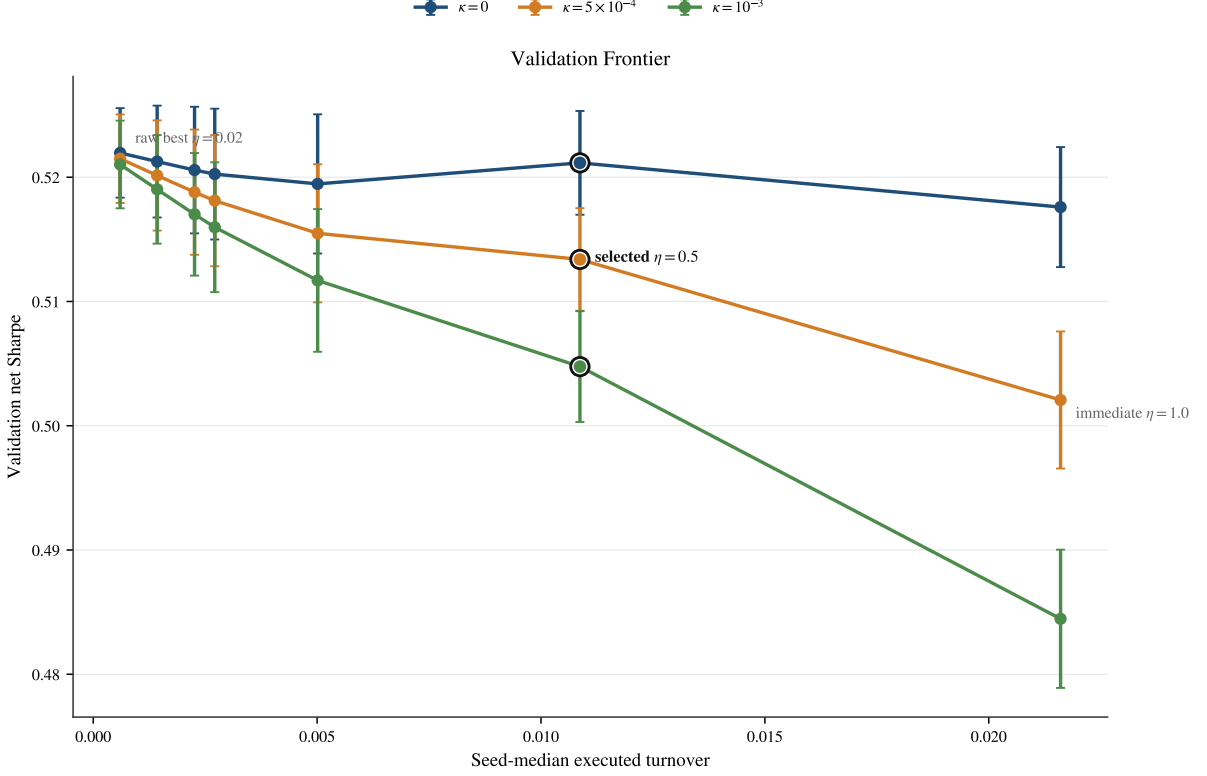


Figure 1: Validation frontier on the effective validation dates 2022-02-15 to 2023-12-29 within the 2022–2023 split. Each line corresponds to one transaction-cost regime $\kappa \in \{0, 5 \times 10^{-4}, 10^{-3}\}$, with vertical bars showing half-IQR of seedwise net Sharpe. The selected operating point $\eta = 0.5$ is highlighted, while the immediate-execution boundary $\eta = 1.0$ and the raw best tiny- η point are annotated only once to keep the frontier readable at print scale.

ing two-sided Wilcoxon signed-rank p -values are 1.95×10^{-3} in both regimes, and the bootstrap 95% confidence intervals for the paired median ΔSharpe remain strictly positive: $[0.0073, 0.0129]$ at $\kappa = 5 \times 10^{-4}$ and $[0.0167, 0.0233]$ at $\kappa = 10^{-3}$. Median executed turnover falls from 0.02200 to 0.01095 in all three reported rows, roughly halving executed trading intensity.

At $\kappa = 0$, the picture is negligible. The paired median Sharpe delta is -0.00025 with a 50% win-rate, sign-test $p = 0.623$, Wilcoxon $p = 0.770$, and a bootstrap 95% confidence interval of $[-0.0022, 0.0028]$ for the paired median ΔSharpe . This is consistent with little difference in the realized trading path once explicit transaction costs are removed, and it does not support a strong regularization or training-superiority interpretation. The regime contrast is therefore clear: evidence is strong for cost-aligned improvement when $\kappa > 0$, and absent when $\kappa = 0$. Because the first two Sharpe columns report marginal arm medians while ΔSharpe reports the median of seed-wise paired differences, the paired median need not equal the arithmetic difference between the two

marginal medians. Because the executed holdings path is unchanged when only κ varies, the $\kappa = 0$ row also reveals the held-out gross-path comparison between $\eta = 0.5$ and $\eta = 1.0$. Appendix Table 9 uses that identity to report the gross-Sharpe and gross-CAGR comparison explicitly alongside turnover and tracking diagnostics.

7.4 Trace-based diagnostics

The trace-based diagnostics in Table 4 are more informative than the original signed mean-gap table. The selected operating point keeps the median turnover ratio $\overline{\text{TO}}^{\text{tgt}}/\overline{\text{TO}}^{\text{exec}}$ essentially fixed at 2.0, which is consistent with the partial-execution mechanism of $\eta = 0.5$. Average tracking discrepancy is modest but non-zero at about 0.00259 in all reported rows. Once the target trace is computed from the target weights rather than by reusing the executed gross return, the $\kappa = 0$ row no longer collapses to zero: median mean absolute return gap is 3.00×10^{-5} and median final equity gap is 6.72×10^{-4} even without transaction costs. Positive

Table 3: Held-out test comparison between the validation-selected operating point $\eta = 0.5$ and the immediate-execution baseline $\eta = 1.0$ on the effective held-out dates 2024-02-14 to 2025-12-31 within the 2024–2025 split. Main values are medians across 10 paired seeds; bracketed terms are IQRs.

κ	Selected Sharpe [IQR]	Baseline Sharpe [IQR]	Paired Δ Sharpe [IQR]	Win-rate	Exact sign p	Paired Δ CAGR [IQR]	Paired $\Delta\overline{\text{TO}}^{\text{exec}}$
0	1.1554 [0.0138]	1.1542 [0.0148]	-0.0002 [0.0028]	0.50	0.6230	-0.0000 [0.0006]	-0.01105
5×10^{-4}	1.1447 [0.0156]	1.1340 [0.0165]	+0.0105 [0.0051]	1.00	0.0010	+0.0017 [0.0009]	-0.01105
10^{-3}	1.1340 [0.0173]	1.1127 [0.0159]	+0.0213 [0.0046]	1.00	0.0010	+0.0035 [0.0009]	-0.01105

Note: “Selected Sharpe” and “Baseline Sharpe” are marginal medians within each arm. “Paired Δ Sharpe” and “Paired Δ CAGR” are medians of seed-wise paired differences and therefore need not equal the arithmetic difference of the two marginal medians. Exact sign p -values are one-sided. In the positive-cost rows, the paired Sharpe delta has two-sided Wilcoxon $p = 0.00195$ and bootstrap 95% intervals strictly above zero. The paired CAGR delta shows the same pattern, with Wilcoxon $p = 0.00195$ and bootstrap 95% intervals [0.00119, 0.00218] and [0.00271, 0.00385]. At $\kappa = 0$, both paired deltas have Wilcoxon $p = 0.76953$, and both bootstrap intervals cross zero.

costs then widen the realized-path divergence, with median final equity gap increasing from 0.00369 at $\kappa = 5 \times 10^{-4}$ to 0.00739 at $\kappa = 10^{-3}$, while median max daily gap rises from 3.04×10^{-4} to 3.18×10^{-4} .

These trace-based diagnostics are still small in absolute magnitude, which is consistent with the fact that the main benefit of execution control is not to create a radically different strategy, but to make the realized path cost-aligned and less turnover-intensive. The cleaner message is therefore about turnover, tracking, and small-but-measurable realized-path divergence, not about large standalone mispricing of the target path.

7.5 Rolling-window robustness: frontier persists, selected-point transfer is window-sensitive

To test whether the canonical 2022–2025 result is specific to that recent window, we repeated the same frozen-policy evaluation on two additional rolling splits: Split A trains on 2010–2017, validates on 2018–2019, and tests on 2020–2021; Split B trains on 2010–2019, validates on 2020–2021, and tests on 2022–2023; Split C is the canonical 2010–2021 / 2022–2023 / 2024–2025 split. As a selected- η transfer test, the result is mixed. Under the locked 95% largest-qualifying rule, Splits A and B both keep $\eta = 1.0$ because the immediate-execution validation score still retains 98.17% and 97.94% of the split-wise best score, respectively. In those two windows, the selected-vs.-baseline held-out delta is therefore identically zero by construction. Split C retains the original selected point $\eta = 0.5$ and the positive-cost gains reported above. These are different objects: selected-point transfer asks whether one locked operating point survives unchanged across windows, whereas frontier robustness asks whether the held-out positive-cost frontier still contains beneficial interior points somewhere along the fixed η grid.

Read as a frontier-robustness diagnostic, however, the picture is stronger. In all three rolling windows, the held-out positive-cost frontier still contains interior execution rates $\eta < 1$ that improve net Sharpe and reduce executed turnover relative to $\eta = 1.0$. Using the best interior held-out point only as a post-selection diagnostic, Split A improves median net Sharpe by +0.0082 at $\kappa = 5 \times 10^{-4}$ and +0.0165 at $\kappa = 10^{-3}$ while lowering median executed turnover by 0.01936; Split B improves median net Sharpe by +0.0163 and +0.0308 while lowering median executed turnover by 0.01879; and Split C improves median net Sharpe by +0.0231 and +0.0439 while lowering median executed turnover by 0.02141. The rolling-window conclusion is therefore not that

one validation-selected operating point transfers unchanged in every window, but that under positive costs the held-out execution frontier itself remains interior across windows while the conservative selector can still remain at $\eta = 1.0$ in some regimes. Figure 2 makes this distinction visual: Split A and Split B retain clear interior frontiers even though their locked selected points remain at the immediate-execution boundary.

7.6 Denser friction grid: the frontier steepens as κ grows

We next densify the friction axis on the canonical split by adding $\kappa \in \{2 \times 10^{-4}, 2 \times 10^{-3}\}$ while keeping the frozen policy, the η grid, and the validation-first protocol unchanged. This expansion is diagnostic: it does not retrain any policy and it does not replace the main selected-point result. Two findings are useful. First, the global validation selector remains stable. When positive costs are expanded to $\{2 \times 10^{-4}, 5 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$, the same locked rule still selects $\eta = 0.5$. Relative to immediate execution, that global selected operating point improves paired median net Sharpe by +0.0041, +0.0105, +0.0213, and +0.0409 as κ increases across those four positive-cost regimes, while reducing median executed turnover by 0.01105 in each row.

Second, the full held-out frontier becomes progressively more favorable to interior execution rates as friction grows. Using the best interior point only as a post-selection diagnostic, the paired median net-Sharpe gain versus $\eta = 1.0$ rises monotonically from +0.0113 at $\kappa = 2 \times 10^{-4}$ to +0.0230 at $\kappa = 5 \times 10^{-4}$, +0.0424 at $\kappa = 10^{-3}$, and +0.0761 at $\kappa = 2 \times 10^{-3}$; the corresponding reduction in median executed turnover is essentially constant at about 0.02143 because the comparison is always against the same immediate-execution baseline. A per- κ diagnostic selector tells the same story from the other direction: it remains at $\eta = 1.0$ for $\kappa = 2 \times 10^{-4}$ and 5×10^{-4} , moves to $\eta = 0.5$ at $\kappa = 10^{-3}$, and tightens further to $\eta = 0.2$ at $\kappa = 2 \times 10^{-3}$. The friction-grid expansion therefore supports a simple interpretation: the execution frontier is not only real, but increasingly consequential as proportional trading frictions rise. Figure 3 summarizes this tightening pattern across the dense friction grid.

7.7 Eta-aligned retraining

The canonical evidence uses the frozen-policy protocol, but it is still useful to ask whether the positive-cost sign survives retraining when the policy is trained directly under the selected execution timescale. To test that question without opening a full retraining grid, we reran the canonical U27 experiment

Table 4: Trace-based diagnostics for the validation-selected operating point $\eta = 0.5$ on the effective held-out dates, median across 10 paired seeds. Gap columns compare the executed net path against the hypothetical target-weight net path.

κ	$\overline{TO}^{exec}$	$\overline{TO}^{tgt}$	$\overline{TO}^{tgt}/\overline{TO}^{exec}$	Tracking	Mean return gap	Final equity gap	Max daily gap
0	0.01095	0.02190	2.00	0.00259	3.00e-05	0.00067	2.90e-04
5×10^{-4}	0.01095	0.02190	2.00	0.00259	3.04e-05	0.00369	3.04e-04
10^{-3}	0.01095	0.02190	2.00	0.00259	3.15e-05	0.00739	3.18e-04

with the same controller family, the same split, the same cost scale, and the same signal snapshot, but with both training and evaluation aligned to $\eta = 0.5$. The resulting comparison is again made against the immediate-execution baseline $\eta = 1.0$ under the same retrained checkpoints.

The sign survives. On validation, the retrained $\eta = 0.5$ arm raises median Sharpe from 0.5184 to 0.5215 at $\kappa = 5 \times 10^{-4}$ and from 0.5052 to 0.5157 at $\kappa = 10^{-3}$, while reducing median executed turnover from 0.01721 to 0.00870. On the held-out test, the same retrained comparison raises median Sharpe from 1.1513 to 1.1600 at $\kappa = 5 \times 10^{-4}$ and from 1.1362 to 1.1522 at $\kappa = 10^{-3}$, again with executed turnover reduced from 0.01735 to 0.00880. The positive-cost mean of the seed-median Sharpe therefore moves from 1.1437 to 1.1561 on the held-out test, and all 5/5 retrained seeds improve in both positive-cost regimes. At $\kappa = 0$, by contrast, the held-out Sharpe difference is only +0.0013. Appendix Table 10 and Figure 6 summarize this check.

7.8 U36 replication

We next test whether the execution frontier is tied too closely to the original 27-name basket by rerunning the frozen-policy protocol on a nearby 36-name sector-balanced U.S. large-cap benchmark. This auxiliary universe keeps the same daily timing, controller family, signal snapshot, and evaluation grid, but changes the asset basket and increases the portfolio dimension. The goal is not to claim large- N scaling from a single auxiliary run; it is to test whether the same execution-accounting pattern reappears outside the original U27 basket.

It does. On the U36 validation split, the raw best positive-

cost score again occurs at the smallest interior point $\eta = 0.02$, but the locked 95% largest-qualifying rule again selects the more conservative interior point $\eta = 0.5$. The validation-selected point retains 96.78% of the best positive-cost validation score while reducing median executed turnover from 0.02224 to 0.01087. On the held-out U36 test, the selected $\eta = 0.5$ point improves median Sharpe from 0.9608 to 0.9721 at $\kappa = 5 \times 10^{-4}$ and from 0.9401 to 0.9636 at $\kappa = 10^{-3}$, while turnover falls from 0.02161 to 0.01061. The raw best interior point remains smaller ($\eta = 0.02$), but the selected point still retains 98.12% of the held-out best positive-cost score, which is consistent with the same conservative-selector interpretation used on the canonical benchmark. Appendix Table 11 and Figure 6 summarize the replication.

7.9 CC-TA-LBIP comparator

The heuristic baselines above use deliberately simple portfolio rules. To compare the RL results against a stronger classical optimizer under the same state information, we next evaluate CC-TA-LBIP, the cost-calibrated information-parity convex baseline described in Section 6. CC-TA-LBIP fits the same linear ridge forecast map on the same 918-dimensional state as the RL controller and then internalizes turnover regularization directly inside the target optimizer rather than through a second PEC layer. The validation-only sweep over $c \in \{0, 2000, 3000, 4000, 6000, 8000, 12000\}$ selects $c = 3000$.

This auxiliary comparator is materially stronger and more consistent than the earlier free- τ linear prototype. On validation, the selected $c = 3000$ point improves net Sharpe

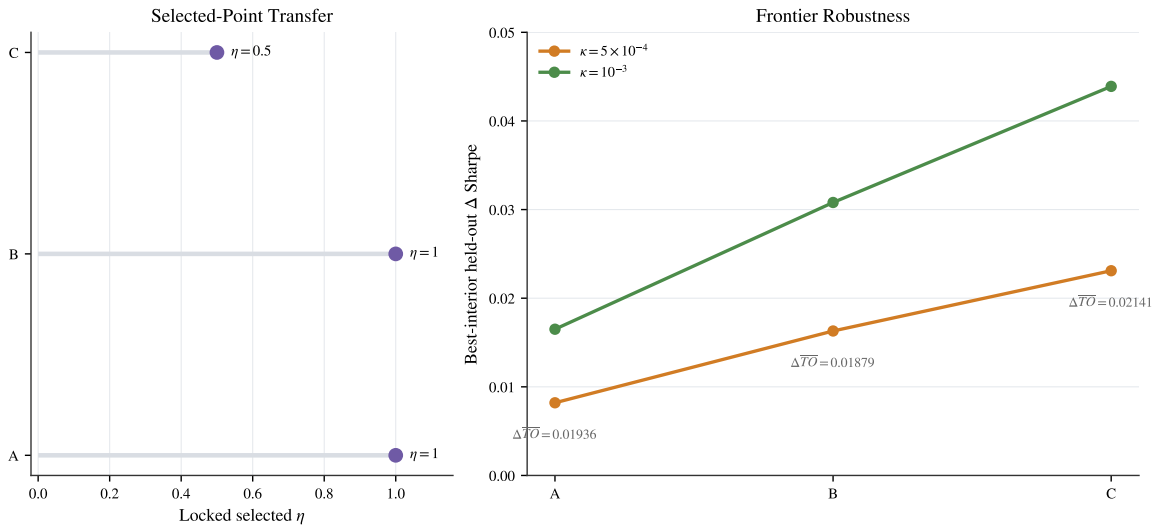


Figure 2: Rolling-window summary with the two relevant objects separated. Left: locked selected-point transfer, shown as the selected η on each split. Right: frontier robustness, shown as the best-interior held-out paired Δ Sharpe at the two positive-cost levels together with the corresponding reduction in executed turnover. The figure makes the paper’s reading explicit: Split C selects an interior point, whereas Splits A and B remain at $\eta = 1.0$ under the conservative selector even though interior positive-cost improvements still exist within the fixed grid.

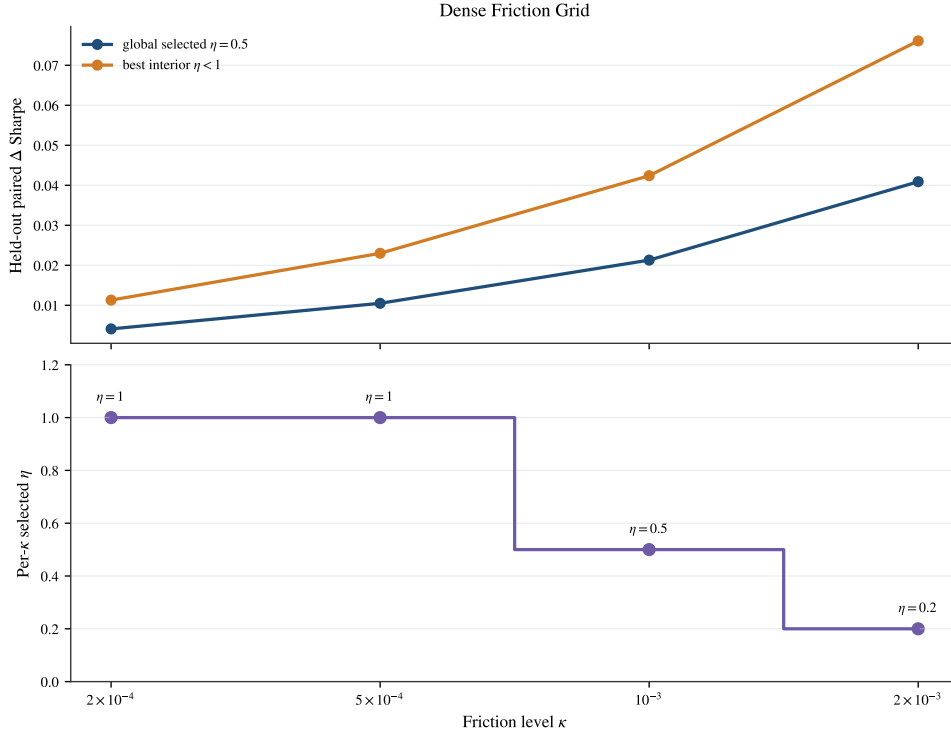


Figure 3: Dense canonical friction-grid summary relative to immediate execution $\eta = 1.0$. The top panel compares the held-out paired Δ Sharpe of the global selected operating point $\eta = 0.5$ against the best interior diagnostic point as friction increases from 2×10^{-4} to 2×10^{-3} . The bottom panel records the per- κ qualifying selector, which stays at the immediate-execution boundary for smaller frictions and moves inward only once costs become large enough.

from -0.5047 to 0.3435 at $\kappa = 5 \times 10^{-4}$ and from -1.5017 to 0.2896 at $\kappa = 10^{-3}$ relative to the unregularized $c = 0$ baseline, while executed turnover falls from 1.3130 to 0.1077 . On the held-out 2024–2025 test, the same validation-selected point improves net Sharpe from 0.0661 to 1.0199 at $\kappa = 5 \times 10^{-4}$ and from -1.0529 to 0.9679 at $\kappa = 10^{-3}$, with turnover reduced from 1.2903 to 0.1189 . Because the integrated turnover term vanishes exactly when $\kappa = 0$, the zero-cost comparison is now identically unchanged: at $\kappa = 0$, Sharpe remains 1.1875 under both $c = 0$ and $c = 3000$. We therefore interpret CC-TA-LBIP as a cleaner external classical comparator under information parity: it shows that a transaction-cost-aware linear-convex optimizer can also benefit materially from cost-calibrated turnover control, while leaving the paper’s main frozen-policy RL identification result unchanged. Appendix Table 12 reports the detailed rows.

Taken together, the three scope checks point in the same direction. The retraining comparison preserves the positive-cost sign with almost no zero-cost movement, the U36 rerun reproduces the same interior execution pattern on a nearby large-cap basket, and the CC-TA-LBIP comparator shows that friction-calibrated turnover control also matters for a stronger classical optimizer under information parity.

7.10 Adaptive η_t schedules are out of scope

Adaptive schedules are left out of the present version. The goal of this paper is to isolate the execution effect under a single frozen control model; adding rule-based η_t schedules would mix execution design with schedule-selection effects. We therefore treat adaptive execution as future work and use the fixed- η frontier as the main empirical object.

8 Discussion

8.1 Mechanistic interpretation

The key result is implementation, not prediction. On the canonical 2024–2025 split, replacing immediate execution with the validation-selected interior point $\eta = 0.5$ roughly halves median executed turnover and improves net Sharpe when costs are positive. Appendix Table 9 shows that the corresponding gross-path change is negligible to slightly negative and does not vary with κ , because κ changes only cost deductions on the same executed path. The clearest reading is therefore simple: the gain comes from lower realized turnover and lower realized cost, not from a better forecasting model or additional alpha. The scope checks point in the same direction. After η -aligned retraining on U27, the held-out positive-cost gains remain $+0.0087$ and $+0.0160$ at the two positive-cost values while turnover again falls by about half. On the second U36 universe, the selected interior point delivers held-out gains of $+0.0113$ and $+0.0236$ with a turnover reduction of about 50.9%. Those checks do not replace the canonical frozen-policy result, but they make the implementation interpretation harder to dismiss as either a frozen-only or one-basket artifact. Appendix Figure 6 visualizes the corresponding held-out auxiliary frontiers.

8.2 When execution helps

The evidence also clarifies when execution control matters most. First, the gains are concentrated in positive-cost regimes: on the canonical split, paired median net-Sharpe improvements are $+0.0105$ at $\kappa = 5 \times 10^{-4}$ and $+0.0213$ at $\kappa = 10^{-3}$. The same regime contrast appears in the aux-

iliary checks: the retrained U27 comparison is essentially flat at $\kappa = 0$ and positive at both positive-cost values, while the U36 replication also turns from negligible at $\kappa = 0$ to clearly positive once costs are introduced. Second, the selected interior operating point achieves these gains with a substantial reduction in executed turnover and only modest target–executed mismatch. Third, on the denser canonical friction grid, the best interior held-out gain steepens monotonically with κ , indicating that the execution frontier becomes more consequential as frictions rise. Taken together, these patterns support a friction-sensitive implementation view rather than a claim that smaller η is universally better.

8.3 Rolling-window reading

Across windows, frontier existence is more stable than selected-point transfer: two additional splits keep the locked selector at $\eta = 1.0$, yet all three windows still contain interior positive-cost points that improve net Sharpe while lowering executed turnover relative to $\eta = 1.0$.

8.4 External context

The matched heuristic baselines are contextual only. On this held-out window, passive exposure was strong: buy-and-hold equal weight exceeds the selected RL arm under aligned executed-path accounting. The auxiliary CC-TA-LBIP comparator adds a stronger classical benchmark under the same information set, but it is still a benchmark rather than a second identification experiment. Its turnover anchor is calibrated directly to κ , so the zero-cost row is identical to the unregularized $c = 0$ baseline and the positive-cost gains are easier to read as friction-sensitive implementation effects rather than free smoothing. The present paper therefore still does not establish that the underlying RL policy dominates passive or linear-convex baselines. Its contribution is narrower: for a fixed learned strategy, execution-aware implementation materially changes realized net performance under frictions. Within that narrower question, the selected operating point is competitive against several higher-turnover comparators, mixed against daily-rebalanced equal weight and long-only mean-variance, and weaker than buy-and-hold equal weight.

9 Limitations

9.1 Frozen-policy scope

The canonical identification result is still frozen-policy. An auxiliary U27 check shows that the positive-cost sign survives one η -aligned retraining comparison at $\eta = 0.5$, but that is still a narrow retraining exercise. The paper therefore does not establish training-time optimality, retraining-frequency effects, or how the frontier would move under a broader retraining grid.

9.2 Universe and realism

The data remain fixed large-cap U.S. equity snapshots rather than point-in-time constituent histories. We now include an auxiliary 36-name benchmark in addition to the canonical 27-name benchmark, which reduces one-basket dependence,

but both datasets still share the same broad market, the same daily abstraction, and the same survivorship-related caveats. Absolute performance levels and external benchmark rankings may therefore be optimistic even though the within-policy η comparison is less exposed.

9.3 Strategy class restriction

The study is limited to a no-cash, long-only, fully invested simplex with daily close-to-close execution. The conclusions therefore do not yet cover cash holdings, leverage, short positions, or intraday microstructure.

9.4 Generalization

The evidence is now stronger than a single-basket study, but it is still limited to one market, one controller family, two nearby large-cap universes, and a narrow contextual comparator set. The new CC-TA-LBIP result broadens the non-RL comparison class under information parity, but it is still one deterministic linear-convex comparator rather than an exhaustive family of turnover-aware convex optimizers. The rolling-window diagnostic, the auxiliary retraining check, the second-universe replication, and the CC-TA-LBIP comparator all support the same broad importance of execution design under frictions, but they are not yet evidence of broad cross-market generalization, policy-family invariance, or benchmark dominance.

10 Conclusion

We propose an execution-aware portfolio RL formulation that separates target and executed portfolios and attaches both transaction-cost accounting and primary evaluation to the realized executed path.

Empirically, the validation-selected operating point is $\eta = 0.5$. On the canonical split, it roughly halves executed turnover and improves net Sharpe in the positive-cost regimes. Across rolling windows, the positive-cost execution frontier remains interior. Two auxiliary checks strengthen that reading: the same sign survives an η -aligned retraining comparison on U27, and the frontier reappears on a second 36-name large-cap universe.

The next steps are to broaden the retraining grid beyond a single η -aligned comparison, extend the study to larger point-in-time universes and additional markets, and test richer execution schedules on top of the same locked accounting protocol.

A Reproducibility Checklist

Table 5 collects the locked artifacts and protocol choices needed to rebuild the paper tables. The point of this appendix is not just that the experiment was rerun, but that the data snapshot, split logic, selection rule, and generated traces are all tied to concrete archived paths.

Table 5: Locked protocol, artifact paths, and rebuild signatures for the canonical paper run.

Locked item	Canonical value
Canonical rebuild root	paper_rebuild_20260324T065755Z.
Processed data snapshot	Fixed cache data/processed_u27, with returns materialized from data/processed_u27/returns.parquet.
Frozen config bundle	frozen_protocol/paper_v3/snapshot_control.yaml, repro/rebuilds/paper_rebuild_20260324T065755Z/configs/validation_eta.yaml, and repro/rebuilds/paper_rebuild_20260324T065755Z/configs/final_eta.yaml.
Seed set	$\{0, 1, \dots, 9\}$, fixed before evaluation.
Split boundaries	Train 2010–2021, validation 2022–2023, held-out test 2024–2025; realized validation/test traces begin on 2022-02-15 and 2024-02-14 because of 30-day rolling features.
Primary selection rule	Choose the largest η whose validation score is at least 95% of the best positive-cost score, where the score is the mean of per- κ median sharpe_net_lin over $\kappa \in \{5 \times 10^{-4}, 10^{-3}\}$.
Primary evaluation objects	Executed-path net Sharpe and executed turnover are primary; target-path quantities are archived as diagnostics only.
Per-run archives	Each evaluation arm stores cmd.txt, config.yaml, metrics.csv, trace.parquet, env_signature.json, and model_resolved_path.txt.
Manifests and materialization metadata	repro/manifests/paper_artifact_manifest.json, repro/manifests/figure_manifest.json, and repro/rebuilds/paper_rebuild_20260324T065755Z/configs/materialization_meta.json.
Environment signatures	Per-seed env_signature.json files in the evaluation directories, plus env_signature_hash, package versions, and config hashes in run_metadata*.json.
Hard-locked equations	Core definitions are frozen in (2)–(6).

A.1 Stored trace and metric fields

The daily executed-path traces are stored as `trace.parquet` files and contain the fields used to regenerate both the main metrics and the appendix diagnostics. In the canonical rebuild, the archived frame includes:

- realized-path returns and costs: `portfolio_return`, `reward`, `cost`, `net_return_lin`, `net_return_exp`,
- target-path diagnostics: `portfolio_return_target`, `cost_target`, `net_return_lin_target`,
- turnover and control fields: `turnover_exec`, `turnover_target`, `eta_t`, `lambda_t`,
- tracking and safety fields: `tracking_error_l2`, `collapse_flag`, `collapse_reason`,
- cumulative paths: `equity_gross`, `equity_gross_target`, `equity_net_lin`, `equity_net_lin_target`,
- run identifiers: `date`, `eval_id`, `run_id`, `model_type`, and `seed`.

The arm-level `metrics.csv` files store the corresponding aggregate fields used in the tables, including `sharpe_net_lin`, `cagr`, `maxdd`, `avg_turnover_exec`, `avg_turnover_target`, `tracking_error_l2_mean`, `collapse_flag_any`, and the canonical `trace.path`.

B Data and Protocol Clarifications

B.1 Fixed universe, cache, and adjusted-close data

The paper rebuild uses a cache-only paper-mode snapshot built from daily adjusted close prices retrieved via `yfinance`. Adjusted close is used so that stock splits and dividend adjustments are reflected in the return series rather than treated as spurious price jumps. The processed cache for the locked run is the fixed directory `data/processed_u27`, and the frozen training config uses a fixed-list universe with 27 assets:

AAPL, AMGN, AXP, BA, CAT, CRM, CSCO, CVX, DIS, GS, HD, HON, IBM, INTC, JNJ, JPM, KO, MCD, MMM, MRK, MSFT, NKE, PG, UNH, V, VZ, WMT

This is a reproducible fixed 27-name snapshot, not a historical point-in-time constituent reconstruction. Legacy project naming may still contain “Dow30”, but the locked paper rebuild uses the fixed 27-name universe listed above. The empirical protocol should therefore be interpreted as a controlled benchmark study with survivorship-related limitations, not as a deployment-grade backtest free of constituent-selection bias.

B.2 Decision timing and return interval

The backtest uses a close-to-close convention. At decision time t :

1. the state s_t is built only from information available up to close t ,
2. the policy outputs the target portfolio w_t^{tgt} ,
3. the execution controller forms the executed portfolio w_t^{exec} via (2),
4. transaction cost is charged on executed turnover $\text{TO}_t^{\text{exec}}$ via (5),
5. the realized return applied to the portfolio is the next-period close-to-close return over $(t, t + 1]$.

Thus the paper uses the convention “decide at close t , then hold w_t^{exec} over $(t, t + 1]$.” This ordering is important because both return realization and cost accounting are attached to the executed portfolio rather than to the raw target action.

B.3 Anti-lookahead timing schematic

The timing logic can be summarized as

$$\text{state through close } t \rightarrow w_t^{\text{tgt}} \rightarrow w_t^{\text{exec}} \rightarrow \kappa \text{TO}_t^{\text{exec}} \rightarrow \text{hold over } (t, t + 1] \rightarrow R_{t+1}.$$

This schematic is the implementation counterpart of the formal definitions in Section 3: the target portfolio is proposed at time t , the execution layer forms the realized holdings, costs are charged on executed turnover, and only then is the next-period return realized.

B.4 Frozen split and model-selection lock

The locked rebuild materializes the experiment configuration before evaluation. The train, validation, and held-out test windows are frozen as 2010–2021, 2022–2023, and 2024–2025, respectively. Because the state uses 30-day rolling features, the realized validation/test traces begin on 2022-02-15 and 2024-02-14 inside those split boundaries. The selected signal snapshot is also frozen before validation and test evaluation; in the rebuilt run, the frozen signal list is `reversal_5d` and `short_term_reversal`. The η grid is fixed *a priori*, validation selects one operating point, and the test window is then used only for final evaluation of that selected operating point and matched comparators. Internal forward checks for 2026 were used only for engineering sanity and are not included in the paper tables or in η selection.

B.5 Constraint set and omitted assets

All experiments use a long-only, fully invested simplex. In particular, portfolio weights are constrained to be non-negative and to sum to one each period. The present study does not include an explicit cash asset, leverage, or short positions. This means that the execution controller studies the effect of partial adjustment within a risky-asset simplex only. The no-cash design is intentional for interpretability, but it also limits the strategy class and should be read as a scope restriction rather than as a claim about the best practical trading interface.

B.6 Backtest scope limitations

The protocol is designed to be reproducible and internally consistent, but its scope remains limited in several ways:

- fixed large-cap U.S. equity snapshots (canonical 27-name benchmark plus one auxiliary 36-name benchmark) rather than point-in-time constituent reconstruction,
- no explicit cash asset,
- long-only fully invested simplex,
- daily close-to-close execution abstraction rather than intraday microstructure modeling,
- canonical frozen-policy identification plus one narrow η -aligned retraining check rather than a broad retraining study.

These limitations do not invalidate the execution-accounting contribution, but they do bound the external claims that should be made from the current experiments.

C Implementation Details

C.1 Observation and action instantiation

Table 6 summarizes the instantiated observation and action interface used by the frozen paper rebuild.

Table 6: Observation and action instantiation for the frozen paper rebuild.

Component	Instantiated value
Data source and cache	Daily adjusted close prices retrieved once via <code>yfinance</code> and then reused in cache-only paper mode.
Universe treatment	Fixed 27-name large-cap U.S. equity snapshot; no point-in-time constituent reconstruction.
Return history block	$L = 30$ daily log-return history flattened in asset order.
Volatility block	$L_v = 30$ rolling per-asset volatility features.
Portfolio-state block	Previous executed weights from the realized holdings at $t - 1$.
Signal block	Two channels: <code>reversal_5d</code> and <code>short_term_reversal</code> .
Observation dimension	$30 \times 27 + 27 + 27 + 2 \times 27 = 918$, matching stored run metadata.
Action space	One real-valued logit per asset in a 27-dimensional box action space.
Simplex mapping	Action clipped to $[-1, 1]$ and mapped to the long-only simplex by a stable softmax with logit scale 10.0.

C.2 Controller schematic

Figure 4 summarizes the frozen controller interface at the level relevant for this paper: the actor proposes target logits, the environment maps them to target weights, the execution layer produces realized holdings, and cost-aligned accounting attaches both turnover and primary evaluation to that realized path.

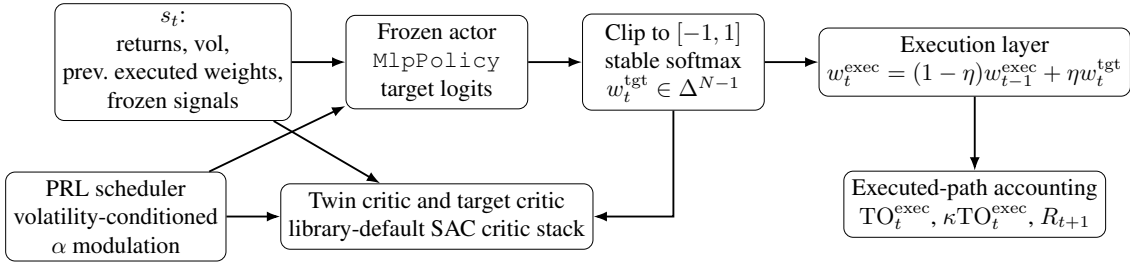


Figure 4: Controller-level schematic for the frozen paper rebuild. The figure is not meant to introduce a novel network design; it makes explicit which parts of the controller are fixed, where the PRL scheduler enters, and where executed-path accounting is attached.

C.3 Frozen controller configuration

Table 7 summarizes what is fixed in the controller architecture, optimization setup, and execution protocol.

C.4 Execution-grid provenance

- The evaluation grid is fixed to $\eta \in \{1.0, 0.5, 0.2, 0.1, 0.082, 0.05, 0.02\}$ before validation and test evaluation.
- The value $\eta = 0.082$ is not a post-hoc insertion: it is the incumbent `rebalance_eta` stored in the frozen control snapshot used to train the reference policy.
- Validation selects one operating point from that fixed grid, and the held-out test split is then reserved for final evaluation only.

C.5 Validation-threshold sensitivity

Table 8 records the small threshold check referenced in the main text. The practical message is that the selected operating point is stable to a moderate tightening of the retained-score threshold: moving from 95% to 97.5% leaves the selected operating point at $\eta = 0.5$, whereas loosening the rule to 90% collapses back to immediate execution.

Table 7: Frozen controller architecture and optimization details.

Component	Locked value
RL backend	Stable-Baselines3 SAC with stored metadata version <code>sb3_version=2.7.1</code> .
Actor backbone	Library-default SB3 SAC <code>MlpPolicy</code> actor; no custom <code>policy_kwargs</code> or explicit <code>net_arch</code> override is stored in the frozen configs or run metadata.
Critic backbone	Library-default SB3 SAC twin-critic stack plus target critic; no custom critic-topology override is stored in the frozen configs or run metadata.
Action parameterization	One scalar logit per asset is emitted by the actor, clipped to $[-1, 1]$, and then mapped to the long-only simplex by the environment-side stable softmax.
Checkpoint reuse	One checkpoint trained once per seed on the training window, then reused across all η arms for that seed.
Execution interface	Policy emits target-allocation logits; the environment maps them to simplex weights and the execution layer applies fixed partial adjustment.
Frozen signals	<code>reversal_5d</code> and <code>short_term_reversal</code> .
Training horizon and cadence	100000 training timesteps per seed, evaluation cadence 5000 steps, checkpoint cadence 50000 steps, and log interval 1000 steps in the frozen control snapshot.
Optimization settings	Shared SAC learning rate 10^{-4} , batch size 256, discount $\gamma = 0.99$, target-update rate $\tau = 0.005$, replay buffer size 100000. No custom optimizer override is specified in the frozen training code, so optimization follows the library SAC defaults under those locked scalar settings.
Entropy handling	The frozen snapshot stores <code>ent_coef=0.2</code> . In PRL-SAC training, SB3 entropy auto-updating is disabled and the scheduler-supplied α enters both the critic target and actor objective, while the evaluation sweep reuses the already-trained frozen checkpoints.
PRL scheduler and penalties	$\alpha_0 = 0.2$, $\beta = 1.0$, $\lambda_v = 2.0$, $\text{bias} = -2.0$, $\alpha_{\min} = 0.01$, $\alpha_{\max} = 1.0$, mid-plasticity multiplier 0.7, and CVaR-penalty gamma 0.3. The scheduler modulates the effective entropy weight from volatility-conditioned observations rather than changing the execution rule during evaluation.
Checkpoint selection rule	Checkpoints are trained on the train split only. Validation selects η , not a different model checkpoint, and the held-out test reuses that same per-seed frozen checkpoint.
Incumbent execution rate	Frozen snapshot <code>rebalance_eta=0.082</code> before evaluation-grid expansion.

Table 8: Sensitivity of the validation selector to the retained-score threshold. The raw-best validation point is $\eta = 0.02$ with normalized score 100%.

Retained-score threshold	Selected η	Normalized score kept	Mean positive-cost tracking
90%	1.0	94.63%	0.00000
95%	0.5	97.66%	0.00257
97.5%	0.5	97.66%	0.00257

C.6 Gross-path mechanism summary

For a fixed executed holdings path, changing κ alters only the cost deduction and leaves gross returns unchanged. The gross comparison between the validation-selected operating point and the immediate-execution baseline can therefore be read directly from the $\kappa = 0$ row of the held-out paired test. Table 9 makes that implication explicit and places the gross-path quantities next to the turnover and tracking diagnostics that explain the positive-cost net benefit.

Table 9: Held-out gross-path summary for the validation-selected operating point $\eta = 0.5$ relative to the immediate-execution baseline $\eta = 1.0$. Gross quantities are read from the $\kappa = 0$ paired comparison because varying κ changes cost accounting but not the executed holdings path. Values are medians across 10 paired seeds; bracketed terms are IQRs.

Quantity	Value
Selected gross Sharpe [IQR]	1.1554 [0.0138]
Baseline gross Sharpe [IQR]	1.1542 [0.0148]
Paired Δ gross Sharpe [IQR]	-0.0002 [0.0028]
Paired Δ gross CAGR [IQR]	-0.0000 [0.0006]
Paired $\Delta \overline{\text{TO}}^{\text{exec}}$	-0.01105
Median target-to-executed turnover ratio $\overline{\text{TO}}^{\text{tgt}} / \overline{\text{TO}}^{\text{exec}}$	2.00
Median tracking error $\ w_t^{\text{exec}} - w_t^{\text{tgt}}\ _2$	0.00259

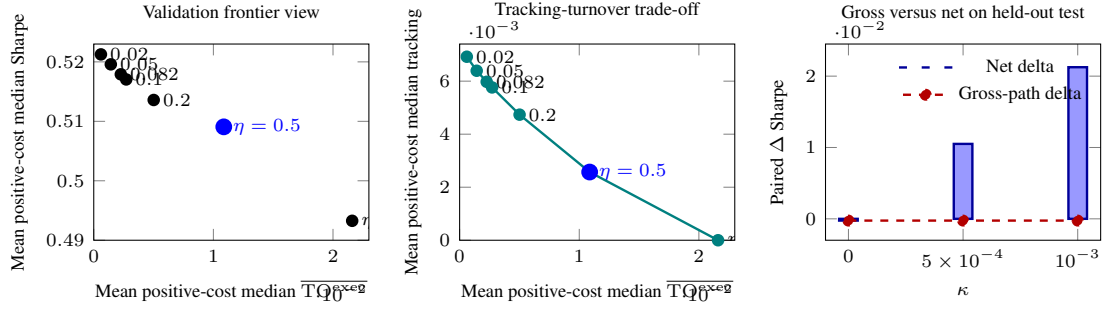


Figure 5: Appendix trade-off view built from already-generated artifacts, with no new experiments. Left: validation positive-cost frontier over the fixed η grid, showing that the locked rule selects the largest qualifying interior point $\eta = 0.5$ rather than the raw best tiny- η point. Middle: lower turnover is achieved by accepting larger target-executed mismatch, but the selected operating point remains far from the extreme low- η end of the tracking curve. Right: on the held-out test, the selected operating point’s net-Sharpe gain grows with friction, while the gross-path Sharpe change stays essentially flat and slightly negative.

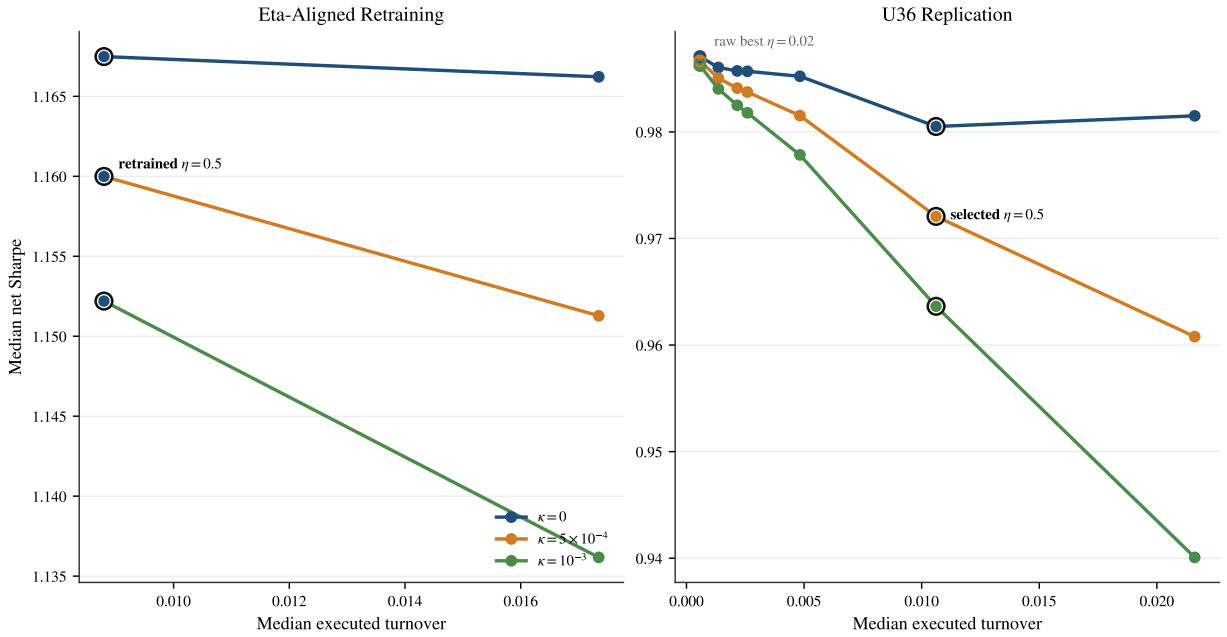


Figure 6: Scope checks for the main identification result. Left: after η -aligned retraining on U27, the interior point $\eta = 0.5$ still cuts turnover roughly in half and remains better in positive-cost regimes while the zero-cost row stays nearly flat. Right: on the U36 benchmark, the same interior operating point is selected again, while the raw best diagnostic point stays at a smaller η . These checks do not replace the main protocol, but they reduce the risk that the main finding is either a frozen-only artifact or a one-basket artifact.

D Auxiliary Detailed Tables

Table 10: η -aligned retraining check on the canonical U27 benchmark. A fresh policy is trained with the same split, controller family, cost scale, and signal snapshot, but with training and evaluation aligned to $\eta = 0.5$. The table then compares the retrained $\eta = 0.5$ arm against the immediate-execution baseline $\eta = 1.0$ under the same retrained checkpoints. Values are medians across 5 paired seeds.

Window	κ	Baseline Sharpe	Retrained $\eta = 0.5$ Sharpe	Δ Sharpe	Baseline TO^{exec}	Retrained TO^{exec}	Seed wins
Validation	0	0.5294	0.5272	-0.0022	0.01721	0.00870	2/5
Validation	5×10^{-4}	0.5184	0.5215	+0.0030	0.01721	0.00870	5/5
Validation	10^{-3}	0.5052	0.5157	+0.0105	0.01721	0.00870	5/5
Held-out test	0	1.1662	1.1675	+0.0013	0.01735	0.00880	3/5
Held-out test	5×10^{-4}	1.1513	1.1600	+0.0087	0.01735	0.00880	5/5
Held-out test	10^{-3}	1.1362	1.1522	+0.0160	0.01735	0.00880	5/5

Note: The held-out positive-cost mean of the seed-median Sharpe moves from 1.1437 under $\eta = 1.0$ to 1.1561 under retrained $\eta = 0.5$. The corresponding turnover reduction is approximately 49.3%.

Table 11: Second-universe replication on a fixed 36-name sector-balanced U.S. large-cap benchmark. The locked validation rule again selects $\eta = 0.5$ even though the raw best positive-cost point is $\eta = 0.02$. Values are medians across 5 seeds.

κ	Immediate $\eta = 1.0$ Sharpe	Selected $\eta = 0.5$ Sharpe	Δ Sharpe	Raw-best interior η	Raw-best interior Sharpe	Turnover ratio ($\eta = 0.5/1.0$)
0	0.9815	0.9805	-0.0010	0.02	0.9871	0.491
5×10^{-4}	0.9608	0.9721	+0.0113	0.02	0.9866	0.491
10^{-3}	0.9401	0.9636	+0.0236	0.02	0.9862	0.491

Note: On validation, the selected point $\eta = 0.5$ keeps 96.78% of the best positive-cost score under the locked 95% largest-qualifying rule. On the held-out test, the selected point keeps 98.12% of the raw best positive-cost score while reducing executed turnover by about 50.9% relative to $\eta = 1.0$.

Table 12: Cost-Calibrated Turnover-Aware Linear Information-Parity Baseline (CC-TA-LBIP) on the canonical U27 split. CC-TA-LBIP uses the same 918-dimensional state as the RL controller, fixes the ridge forecast map at `ridge_alpha = 30`, and tunes only the cost-scaling constant c on validation. The selected operating point is $c = 3000$, the baseline comparator is the unregularized $c = 0$ variant, and the integrated turnover anchor is proportional to the actual friction level κ . Values are from a deterministic single-run comparator rather than a paired multi-seed RL sweep.

Window	κ	Selected c	Selected Sharpe	Baseline Sharpe ($c = 0$)	Δ Sharpe	Selected $\overline{TO}^{\text{exec}}$
Validation	0	3000	0.4939	0.4939	0.0000	1.3130
Validation	5×10^{-4}	3000	0.3435	-0.5047	+0.8482	0.1077
Validation	10^{-3}	3000	0.2896	-1.5017	+1.7913	0.0570
Held-out test	0	3000	1.1875	1.1875	0.0000	1.2903
Held-out test	5×10^{-4}	3000	1.0199	0.0661	+0.9539	0.1189
Held-out test	10^{-3}	3000	0.9679	-1.0529	+2.0208	0.0633

Note: At $\kappa = 0$, the integrated turnover anchor is exactly zero for all c , so the selected and baseline rows are identical by construction. At positive frictions, the selected CC-TA-LBIP point reduces executed turnover by about 91.8% on validation and 90.8% on the held-out test relative to $c = 0$.

E Secondary Matched-Heuristic Context

Table 13 is reported only as contextual external scale, not as identification evidence. It is not used for η selection, not used for validation decisions, and not used for the paper’s core causal interpretation. Under the same effective held-out dates, the same transaction-cost definition $\kappa \cdot \overline{TO}^{\text{exec}}$, the same annualization $\sqrt{252}$, and the same $r_f = 0$ convention, the selected operating point is stronger than inverse-volatility risk parity and minimum-variance, mixed against daily-rebalanced equal weight and long-only mean-variance, and weaker than buy-and-hold equal weight on this window. The table therefore helps calibrate external performance scale under aligned accounting, while leaving the main claim unchanged: the paper studies the within-policy execution differential of a fixed learned strategy.

Table 13: Held-out test comparison between the validation-selected operating point $\eta = 0.5$ and matched heuristic baselines on the effective held-out dates 2024-02-14 to 2025-12-31 within the 2024–2025 split. All quantities use the same executed-path accounting conventions.

κ	Baseline	Baseline Sharpe	Δ Sharpe	Baseline CAGR	Δ CAGR	$\Delta \overline{TO}^{\text{exec}}$
0	Buy-and-hold EW	1.1726	-0.0172	0.1651	-0.0001	+0.01095
0	Daily-rebalanced EW	1.1530	+0.0024	0.1646	+0.0004	+0.00106
0	Inverse-vol RP	1.0772	+0.0782	0.1416	+0.0234	-0.02210
0	Minimum-variance	1.0649	+0.0905	0.1219	+0.0431	-0.04160
0	Mean-variance (long-only)	1.2269	-0.0716	0.1841	-0.0190	-0.08780
5×10^{-4}	Buy-and-hold EW	1.1726	-0.0279	0.1651	-0.0018	+0.01095
5×10^{-4}	Daily-rebalanced EW	1.1441	+0.0005	0.1632	+0.0001	+0.00106
5×10^{-4}	Inverse-vol RP	1.0453	+0.0994	0.1369	+0.0264	-0.02210
5×10^{-4}	Minimum-variance	1.0064	+0.1382	0.1145	+0.0487	-0.04160
5×10^{-4}	Mean-variance (long-only)	1.1420	+0.0027	0.1694	-0.0061	-0.08780
10^{-3}	Buy-and-hold EW	1.1726	-0.0386	0.1651	-0.0035	+0.01095
10^{-3}	Daily-rebalanced EW	1.1353	-0.0013	0.1617	-0.0002	+0.00106
10^{-3}	Inverse-vol RP	1.0133	+0.1206	0.1321	+0.0294	-0.02210
10^{-3}	Minimum-variance	0.9480	+0.1860	0.1072	+0.0544	-0.04160
10^{-3}	Mean-variance (long-only)	1.0570	+0.0770	0.1550	+0.0066	-0.08780

Note: All heuristic rows use the same effective held-out dates 2024-02-14 to 2025-12-31 and the same executed-path accounting conventions. This table is contextual only: it is not part of the identification strategy and is not used for η selection, validation, or the paper’s core causal interpretation. Positive $\Delta \overline{TO}^{\text{exec}}$ means the selected RL arm trades *more* than the comparator; negative values mean it trades less.

F Supplementary Figures

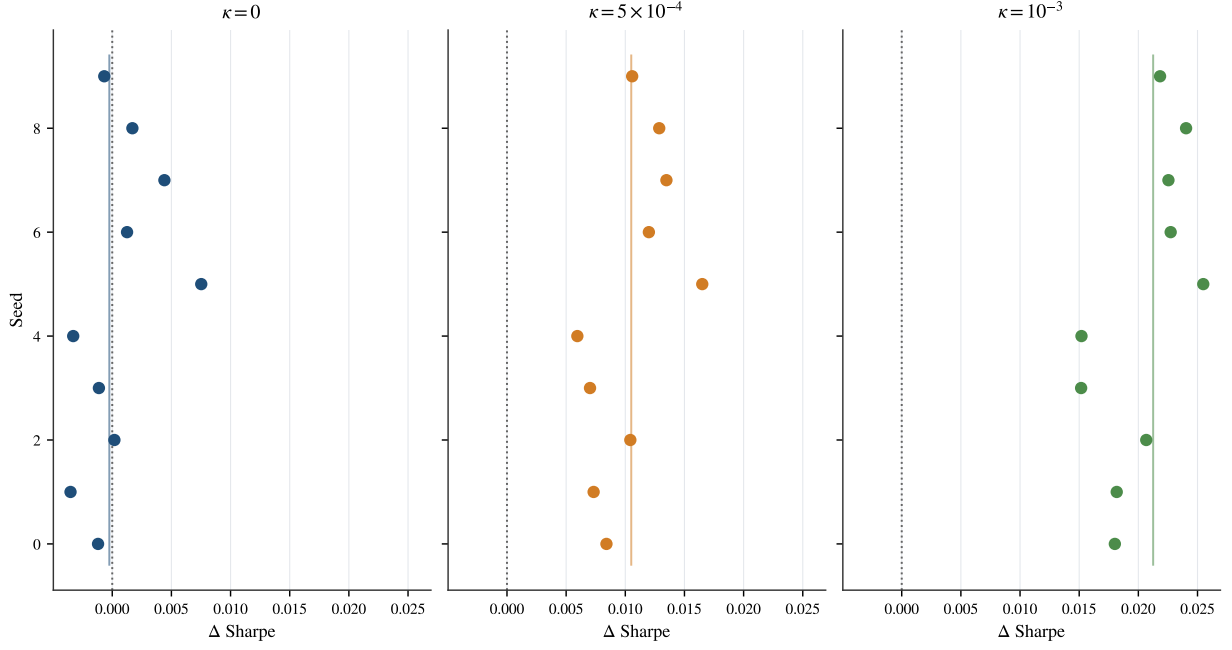


Figure 7: Seed-wise paired Δ net Sharpe on the effective held-out dates 2024-02-14 to 2025-12-31 for the validation-selected operating point $\eta = 0.5$ relative to the immediate-execution baseline $\eta = 1.0$. Each point is one seed-wise difference $\text{Sharpe}_{\eta=0.5} - \text{Sharpe}_{\eta=1.0}$, and the vertical reference line marks zero. All 10 paired seeds are positive in the two positive-cost regimes, matching the 10/10 seed-win result reported in Table 3, whereas the $\kappa = 0$ panel mixes positive and negative deltas.

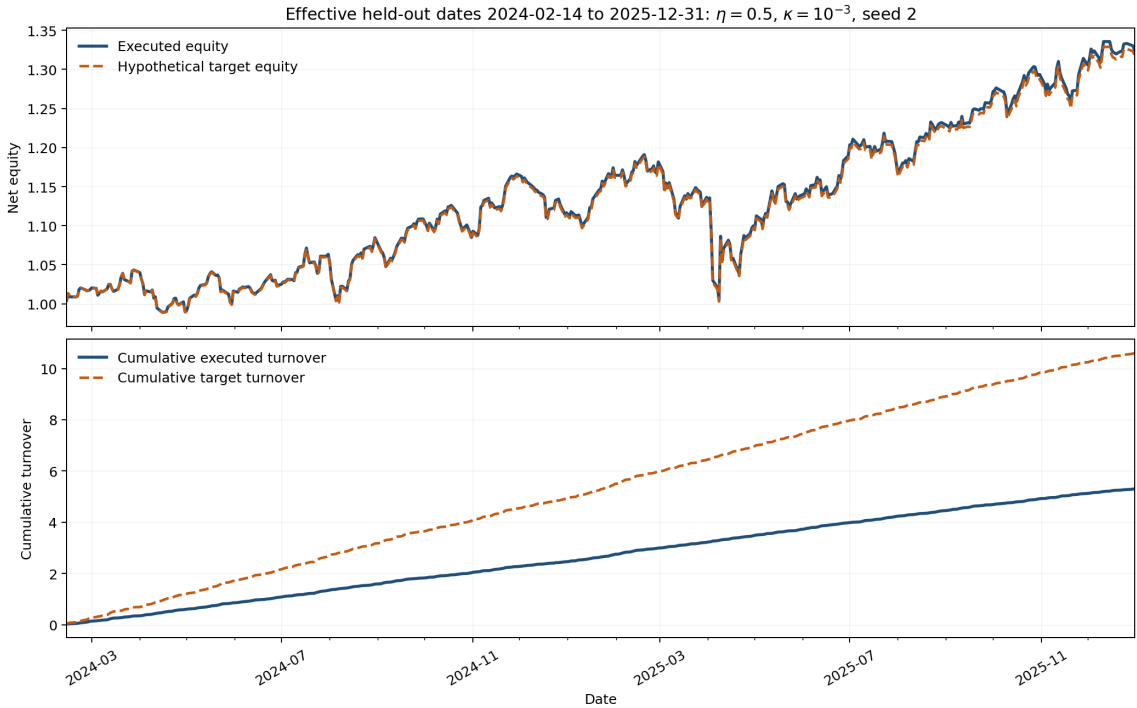


Figure 8: Representative held-out trace for the validation-selected operating point on the effective held-out dates 2024-02-14 to 2025-12-31 within the 2024–2025 split at $\eta = 0.5$, $\kappa = 10^{-3}$, seed 2. The top panel shows executed net equity and hypothetical target net equity, where the target path is the immediate move to the target portfolio used only for diagnostics. The bottom panel shows cumulative executed and target turnover. The representative seed is the selected-arm run whose held-out net Sharpe is closest to the median selected-arm net Sharpe across the 10 paired seeds at $\kappa = 10^{-3}$.

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